

## **KEYSTRA ENERGY GROUP**

### **SOLUTION 1:**

#### **ENTERPRISE HSE PERFORMANCE MANAGEMENT**

**“Protecting Keystra’s workforce through enterprise HSE analytics, performance measurement, and continuous safety improvement”**

#### **BUSINESS QUESTION**

**How can enterprise HSE performance be evaluated to identify safety risks, understand their business impact, and drive continuous improvement?**

##### ***Areas of Investigation>>***

- **Workforce Exposure**

**Where are employees spending the most time exposed to operational activities?**

- **Safety Performance**

**How effectively is the organisation managing workforce safety performance across its operations?**

- **Business Impact & Corrective Actions**

**What business impacts do safety incidents create, and how effective are corrective actions in preventing similar incidents?**

#### **KEY ANALYTICS AREA:**

- **Workforce Exposure Monitoring**
- **Incident & Hazard Analysis**
- **Safety Performance Measurement**
- **Business Impact Assessment**
- **Corrective Action Performance**
- **Enterprise HSE Reporting**

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As an integrated energy company, Keystra Energy Group operates across a diverse portfolio of upstream, midstream, downstream, LNG, and renewable energy assets. These operations rely on thousands of workforce hours every day to maintain production, execute maintenance activities, manage logistics, and deliver critical infrastructure safely and efficiently. In such an operating environment, Health, Safety and Environment (HSE) is more than a compliance requirement, it is a business function that underpins operational continuity, workforce wellbeing, and organisational performance.

As operations grow in scale and complexity, so does the volume of HSE information generated across the organisation. Workforce exposure records, incident reports, hazard observations, and corrective actions are often captured across different operational processes. While each provides valuable information on its own, greater value is created when they are analysed together to provide a complete picture of organisational HSE performance. Without this integrated perspective, it becomes difficult to understand where operational risks are emerging, how safety performance is changing over time, and whether corrective actions are delivering the intended improvements.

To address this need, Keystra's Enterprise Analytics Portfolio begins with the development of an Enterprise HSE Performance Management solution. As the foundation of the portfolio, this solution brings together workforce exposure, incident management, and corrective action management into a single analytical framework designed to monitor HSE performance from operational, strategic, and business perspectives. By combining these critical HSE processes, the organisation is able to measure safety performance, identify areas requiring intervention, evaluate the impact of safety events, and monitor the effectiveness of improvement initiatives across the business.

This report documents the design and implementation of the solution, from the development of the underlying enterprise data model through to the delivery of interactive dashboards that provide meaningful insight into HSE performance. Beyond supporting workforce safety, the solution establishes the first analytical capability within Keystra's Enterprise Analytics Portfolio and provides a foundation for the subsequent enterprise solutions presented throughout this portfolio.

## BUSINESS CONTEXT

Keystra Energy Group operates across upstream, midstream, downstream, LNG, and renewable energy businesses, each characterised by complex operational activities and varying levels of risk. Daily operations involve drilling and production activities, pipeline transportation, plant operations, equipment maintenance, logistics, inspections, and field interventions, all of which require employees and contractors to work in environments where effective HSE management is essential to sustaining safe and reliable operations.

Beyond meeting regulatory obligations, HSE performance plays a significant role in maintaining operational continuity, protecting the workforce, reducing financial losses, and strengthening organisational resilience. As operations continue to expand across multiple assets, departments, and business units, the volume of HSE information generated also increases. Workforce exposure records, incident reports, hazard observations, and corrective action activities collectively provide valuable operational intelligence; however, their true value lies in understanding how these business processes interact to influence overall HSE performance.

To effectively manage these relationships, organisations require an integrated performance management approach that provides visibility across the entire HSE lifecycle; from workforce exposure and safety events to business impacts and corrective actions. This enables leaders to monitor performance consistently, identify areas of elevated risk, evaluate improvement initiatives, and support evidence-based HSE management across the enterprise.

## BUSINESS PROBLEM

Although significant volumes of HSE data are generated during daily operations, much of this information is often analysed independently, resulting in fragmented reporting and limited operational visibility. Incident reports may indicate where safety events have occurred, but they do not explain the level of workforce exposure associated with those events, the operational and financial consequences they create, or whether corrective actions have effectively reduced the likelihood of recurrence.

This fragmented approach makes it difficult to obtain a holistic view of enterprise HSE performance. Consequently, operational risks may remain hidden until incidents occur, improvement efforts may be directed towards the wrong areas, and management may lack the information required to prioritise resources based on actual risk exposure and business impact.

To address these challenges, Keystra requires an integrated HSE performance management solution capable of consolidating workforce exposure, incident management, and corrective action information into a single analytical environment. Such

a solution would enable consistent performance measurement, improved visibility into operational risks, and more effective monitoring of safety improvement initiatives across the organisation.

## **BUSINESS OBJECTIVES AND VALUE**

The primary objective of this solution is to develop an enterprise HSE performance management capability that provides a comprehensive view of workforce safety across Keystra Energy Group. By integrating workforce exposure, incident, and corrective action data into a unified dimensional data warehouse, the solution enables HSE performance to be monitored consistently across business units, departments, operational assets, and work shifts.

Specifically, the solution is designed to:

- \* Monitor workforce exposure across operational activities.
- \* Measure HSE performance using industry recognised indicators such as the Total Recordable Incident Frequency (TRIF) and Lost Time Injury Frequency Rate (LTIFR).
- \* Identify operational areas with elevated safety risk through incident and hazard analysis.
- \* Evaluate the operational and financial impact of workplace incidents.
- \* Monitor corrective action performance to improve accountability and reduce the recurrence of safety incidents.
- \* Deliver enterprise-wide HSE dashboards that support performance monitoring and continuous improvement.

By providing a single source of truth for HSE performance, the solution enables Keystra to move beyond reactive incident reporting towards proactive performance management. It strengthens visibility into workforce safety, improves the prioritisation of risk reduction initiatives, enhances accountability for corrective actions, and supports more effective operational planning. Ultimately, this capability contributes to safer operations, improved business resilience, and a stronger organisational safety culture.

## DATA REQUIREMENTS AND SOURCES

The development of this solution began with identifying the business data required to answer the questions defined in the previous chapter. Since the objective was to evaluate HSE performance across the organisation, the solution required data that could provide visibility into workforce exposure, workplace incidents, hazards, corrective actions, and the organisational structure within which these activities occur. Together, these business processes provide the information needed to understand not only how safely work is being performed, but also the operational and business impact of safety performance across Keystra Energy Group.

Ideally, this type of analysis would be performed using operational HSE data from a real energy company. However, detailed HSE datasets are rarely made publicly available because they contain sensitive operational information, confidential business records, and employee data. Even where limited datasets exist, they often represent only a single aspect of HSE management and do not provide the integrated view required for enterprise performance analysis.

To overcome this challenge, a realistic enterprise HSE dataset was designed and developed specifically for this project using Python. Rather than generating random records, the data was modelled to reflect the structure and operational characteristics of an integrated energy company. Business rules, organisational hierarchies, operational relationships, and realistic distributions were incorporated throughout the data generation process to ensure that the resulting dataset closely represented real world HSE operations.

The final dataset was built around three core business processes that support enterprise HSE performance management: Workforce Exposure Management, Incident and Hazard Management, and Corrective Action Management. These transactional datasets were supported by shared organisational dimensions, providing a consistent analytical foundation for measuring safety performance, evaluating business impact, and monitoring corrective action effectiveness across the organisation.

## DATA PREPARATION AND ENGINEERING

After establishing the project parameters, the next stage involved developing the reference and dimension tables that would form the foundation of the analytical data warehouse. These tables define the key organisational, operational, and HSE entities used throughout the project.

The reference data was created first because the subsequent workforce, exposure, incident, and corrective-action records needed consistent identifiers to connect back to the organisation's structure. Each dimension was therefore assigned unique keys and

reviewed for completeness and consistency before being used in later stages of data generation.

## **DIMENSION TABLES DEVELOPMENT**

### **1. BUSINESS UNIT DIMENSION**

This established the top level structure of Keystra Energy Group. The organisation was divided into five business units representing its integrated energy portfolio:

- \* BU001 — Upstream; represents activities associated with exploration and production, including offshore production operations.
- \* BU002 — Midstream; covers the infrastructure used to process, transport, compress, and move energy products between production and downstream operations.
- \* BU003 — Downstream; represents activities such as refining, fuel distribution, and the handling of petroleum products.
- \* BU004 — LNG; covers the processing, liquefaction, storage, and export of liquefied natural gas.
- \* BU005 — Renewable; represents Keystra's renewable energy portfolio, including solar and wind generation assets.

These business units provide the highest level organisational context for analysing workforce distribution, operational exposure, HSE events, and corrective actions. Unique Business Unit IDs were used throughout the warehouse to ensure that records could be consistently associated with the appropriate part of the organisation.

### **2. DEPARTMENT DIMENSION**

This was created to represent the seven major functional areas within Keystra Energy Group. These departments provide the organisational context needed to understand where employees work and how different functions contribute to operational activities and HSE performance.

The seven departments were: Engineering, Finance and Administration, HSE, Human Resources, Logistics, Maintenance, and Operations. The departments were assigned unique identifiers and later used when developing employee records and modelling differences in workforce exposure and incident risk. Validation checks were performed to confirm that the department identifiers were unique, required values were present, and all seven functional areas were represented.

### **3. ASSET DIMENSION**

The Asset lookup established the operational asset register for Keystra Energy Group. Thirteen assets were created across the organisation's different business units and operating environments.

| ASSET ID | ASSET NAME                 | BUSINESS UNIT | OPERATIONAL CONTEXT                                                                                                                                                                                    |
|----------|----------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AST001   | Fuel Distribution Terminal | Downstream    | Handles the storage, loading, transfer, and distribution of refined petroleum products. Activities involve fuel handling, vehicle movement, loading operations, and material transfer.                 |
| AST002   | Gas Compression Station    | Midstream     | Supports the compression and transportation of natural gas through the midstream network. Activities involve high-pressure systems, rotating equipment, gas handling, and maintenance activities.      |
| AST003   | Gas Processing Plant       | Midstream     | Processes produced natural gas to separate and treat gas components before transportation or further use. It involves process equipment, pressure systems, hydrocarbons, and chemical/process hazards. |
| AST004   | LNG Export Terminal        | LNG           | Supports the storage, handling, and export of liquefied natural gas. It represents a major process facility involving cryogenic systems, hydrocarbons, pressure systems, and marine/export activities. |
| AST005   | Offshore Platform A        | Upstream      | Offshore oil and gas production facility where personnel work around production equipment, hydrocarbons, pressure systems, lifting operations, and marine/offshore conditions.                         |



|        |                        |            |                                                                                                                                                                                                                      |
|--------|------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AST006 | Offshore Platform B    | Upstream   | Offshore production facility supporting upstream activities and involving similar operational risks associated with hydrocarbons, process systems, working at height, lifting, and offshore operations.              |
| AST007 | Offshore Platform C    | Upstream   | Offshore production asset supporting upstream operations, with exposure to production equipment, hydrocarbon processes, pressure systems, lifting activities, and other offshore hazards.                            |
| AST008 | Pipeline Network Alpha | Midstream  | Represents pipeline infrastructure used to transport energy products. Key activities include pipeline operations, inspection, maintenance, pressure management, and field intervention.                              |
| AST009 | Pipeline Network Bravo | Midstream  | A second pipeline network supporting the transportation of energy products and associated inspection, maintenance, monitoring, and operational activities.                                                           |
| AST010 | Refinery Complex       | Downstream | Large-scale downstream facility involved in processing and refining petroleum products. It represents a high-hazard process environment involving hydrocarbons, pressure, heat, chemicals, and process safety risks. |
| AST011 | Solar Farm North       | Renewable  | Renewable energy generation facility based on solar power. Activities include equipment inspection, electrical maintenance, working at height, manual handling, and site operations.                                 |

|        |                  |           |                                                                                                                                                                                                |
|--------|------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AST012 | Solar Farm South | Renewable | Solar generation facility supporting Keystra's renewable energy portfolio, with operational activities centred around solar equipment, electrical systems, maintenance, and site management.   |
| AST013 | Wind Farm        | Renewable | Renewable energy generation facility using wind turbines. Activities include turbine operation and maintenance, electrical systems, working at height, lifting, and specialist technical work. |

This asset structure provides more than simply a list of facilities. It gives the HSE model an operational context. For example, incidents occurring on an offshore platform can be analysed differently from those occurring at a corporate or renewable-energy environment because the nature and severity of the hazards are different. The dimension was validated by checking for unique Asset IDs, complete asset records, and consistency between each asset and its assigned business unit.

#### 4. JOB ROLE DIMENSION

This was developed to provide a more detailed representation of the workforce structure. Rather than assigning employees only to departments, the model included 35 job roles spanning the different functional areas of Keystra Energy Group. The job-role structure allowed employees to be associated with the type of work they perform within their respective departments. This provided additional granularity for workforce analysis and helped the subsequent HSE modelling reflect differences in operational responsibilities.

Job roles were mapped to relevant departments so that roles remained consistent with Keystra's organisational structure. For example, operational roles were associated with Operations, technical roles with Engineering and Maintenance, safety-focused roles with HSE, and workforce-support roles with Human Resources and Finance & Administration.

The dimension was checked for unique role identifiers, valid department associations, and complete role records before being incorporated into the employee dimension.

## 5. SHIFT DIMENSION

The Shift dimension established the three workforce working patterns used throughout the model: Day Shift, Night Shift, and Rotational Shift. These shift types provide an important operational dimension because workforce exposure and HSE risk can vary according to working patterns. In particular, rotational and night-based operations can involve different working conditions from standard daytime activities. It was also validated to ensure that each shift type had a unique identifier and that the reference table contained no missing or duplicate records.

## 6 INCIDENT TYPE DIMENSION

This established the classification framework used to describe HSE events within the organisation. The classification was designed to accommodate both lower-consequence events and high-consequence incidents, allowing the final dataset to support meaningful safety performance analysis. The incident framework included categories such as: Near Miss, First Aid Case, Medical Treatment Case, Restricted Work Case, Lost Time Injury, Fatality, Fire / Explosion, Process Safety Event, Property Damage, and Environmental Incident.

Each incident classification was assigned a unique Incident Type ID. This dimension table became important in the incident generation process because incident types were not treated as equally likely. Their probabilities were adjusted to create a more realistic HSE event distribution, while the resulting classifications were subsequently used to derive consequence indicators such as recordable incidents, LTIs, fatalities, and near misses.

## 7. HAZARD DIMENSION

The Hazard lookup established the six hazard categories and 12 specific hazard names used throughout the HSE model. Rather than treating every hazard as an isolated value, the hazards were organised into broader categories. This made it possible to analyse HSE events at both the detailed hazard level and the higher category level.

Hazard Categories:

### 1. Chemical

The Chemical category represents hazards arising from exposure to hazardous substances, chemicals, and other potentially harmful materials used or encountered during Keystra's operations.

### 2. Energy

The Energy category represents hazardous energy sources that can cause injury or equipment damage if not properly controlled. This includes electrical energy and stored or pressurised energy within operational systems.

### 3. Ergonomic

The Ergonomic category represents risks associated with how work is performed, particularly manual handling, physical workload, repetitive activity, and poor interaction between workers and their working environment.

### 4. Physical

The Physical category covers a broad range of workplace hazards involving the physical environment, equipment, movement, access, and work activities. These hazards are particularly relevant across operational and maintenance environments.

### 5. Process Safety

The Process Safety category represents hazards associated with the loss of containment or uncontrolled process conditions that could result in major incidents, fires, explosions, or significant operational consequences.

This category is particularly important for Keystra because of the presence of hydrocarbon production, gas processing, LNG, refining, and other energy-related operations.

### 6. Transportation

The Transportation category represents hazards associated with vehicles and mobile equipment used to transport personnel, materials, equipment, and products across Keystra's operational environment.

### Specific Hazard Names

The model contained 12 specific hazards distributed across the six categories:

| HAZARD CATEGORY | HAZARD NAME         | RELEVANCE TO KEYSTRA                                                                                                                                |
|-----------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical        | Hazardous chemicals | Represents exposure to harmful chemicals and substances used during processing, maintenance, treatment, cleaning, and other operational activities. |
| Energy          | Electrical energy   | Represents exposure to electrical systems and equipment during operations, maintenance, inspection, and technical work.                             |

|                       |                                      |                                                                                                                                                                                  |
|-----------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Energy</b>         | <b>Pressure/stored energy</b>        | <b>Represents uncontrolled energy stored within pressurised systems, equipment, pipelines, vessels, and other process systems.</b>                                               |
| <b>Ergonomic</b>      | <b>Ergonomic/manual handling</b>     | <b>Represents physical strain and injury risks associated with lifting, carrying, repetitive tasks, awkward postures, and manual work.</b>                                       |
| <b>Physical</b>       | <b>Confined space</b>                | <b>Represents hazards associated with entering or working within enclosed or restricted spaces where access, atmospheric conditions, or rescue may present additional risks.</b> |
| <b>Physical</b>       | <b>Dropped objects</b>               | <b>Represents objects falling from elevated positions or being released during lifting, maintenance, construction, or offshore activities.</b>                                   |
| <b>Physical</b>       | <b>Lifting and material handling</b> | <b>Represents risks associated with cranes, lifting equipment, suspended loads, materials movement, and manual or mechanical handling activities.</b>                            |
| <b>Physical</b>       | <b>Slips, trips and falls</b>        | <b>Represents common workplace hazards caused by uneven surfaces, obstructions, spills, poor housekeeping, and other conditions that can cause falls.</b>                        |
| <b>Physical</b>       | <b>Working at height</b>             | <b>Represents activities performed at elevated locations, including work on platforms, structures, equipment, towers, and other elevated areas.</b>                              |
| <b>Process Safety</b> | <b>Fire/explosion</b>                | <b>Represents uncontrolled combustion or explosion events that can arise from hydrocarbons, ignition sources, process failures, or loss of containment.</b>                      |
| <b>Process Safety</b> | <b>Hydrocarbons release</b>          | <b>Represents unintended releases of hydrocarbons from process equipment, pipelines, storage systems, or production facilities.</b>                                              |

|                |                          |                                                                                                                                                     |
|----------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Transportation | Vehicle/mobile equipment | Represents incidents involving vehicles and mobile equipment used for personnel movement, logistics, material transportation, and field operations. |
|----------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

The hazard structure was particularly important to the incident-generation process because hazards were later assigned according to the department and asset involved in each incident, rather than being distributed uniformly.

For example, employees working in offshore production, gas processing, LNG, or refinery environments were given greater exposure to process-related hazards such as Hydrocarbon Release, Pressure / Stored Energy, and Fire / Explosion. Maintenance activities were more strongly associated with hazards such as Electrical Energy, Lifting & Material Handling, and Manual Handling, while logistics activities had greater relevance to Vehicle / Mobile Equipment and material movement. Lower-risk administrative environments were more strongly associated with Ergonomic / Manual Handling and Slips, Trips & Falls.

This structure allowed the subsequent incident-generation process to reflect the nature of the work and operating environment at Keystra, rather than producing a generic random distribution of hazards.

## 8. CONTRACTOR DIMENSION

This was developed to represent external organisations that support Keystra's operations. Contractors were included because a significant proportion of activities within an integrated energy company can involve specialist external service providers. Contractor activities were designed around areas such as: Maintenance, Engineering, Logistics, Construction, Specialist technical services, and Inspection and related support activities

Each contractor was assigned a unique Contractor ID and associated descriptive information. The contractor structure was subsequently used when modelling contractor involvement in HSE incidents. It was also validated to ensure that contractor identifiers were unique, contractor records were complete, and all contractor references used later in the incident fact table could be traced back to valid contractor records.

## 9. WORKFORCE DIMENSION DEVELOPMENT

With the organisational and operational dimensions established, the next stage focused on developing Dim\_Employee, which provides the workforce foundation of the HSE analytical model. The dimension was designed to capture not only employee identity, but

also organisational assignment, employment characteristics, demographics, experience, reporting structure, and workforce HSE attributes.

The workforce was established at 4,000 employees, with each employee assigned a unique Employee ID. Employees were distributed across business units, departments, assets, job roles, and shifts using probability-based allocations designed to reflect Keystra's operational structure rather than equal distribution. The employee records were further enriched with employment type and status, gender, work location, crew, date of birth, age, nationality, education level, hire date, years of service, years of experience, supervisor ID, and HSE-related attributes such as certification status, training compliance, PPE compliance, medical fitness, and emergency response team status. Logical constraints were applied to maintain consistency between related attributes, including age and date of birth, hire date and date of birth, years of service, and years of experience.

The completed dimension was then validated by checking employee population, unique IDs, organisational and employment distributions, demographic and experience ranges, missing values, duplicate records, and logical date relationships. These checks confirmed that the employee dimension was ready to provide the workforce context required for the subsequent HSE exposure, incident, and corrective-action fact tables.

## 10. LOCATION AND ASSET GEOGRAPHY DEVELOPMENT

Once the workforce structure was in place, the next step was to establish where Keystra's people and operations are located. This was important because HSE performance can vary depending on the environment in which work is carried out. Offshore operations, industrial facilities, corporate offices, and renewable energy sites do not present the same operational conditions or safety exposures.

A Location Dimension was therefore created for Keystra's eight strategic Nigerian locations. Each location was assigned a unique ID and described using its Location Name, State, Country, and Operational Environment. The operational environment captured the categories established for the model: Corporate Office, Offshore/Coastal, Onshore/Industrial, Onshore/Offshore, and Renewable Energy. The operational assets were then mapped to their corresponding locations based on their operating context. This created a consistent relationship between Asset, Business Unit, Location, State, and Operational Environment, allowing future HSE events and workforce exposure to be traced back to the environment in which the work occurred.

The mappings were validated to confirm that every asset had a valid location, all referenced Asset and Location IDs existed, no duplicate or conflicting mappings were introduced, and no required location assignments were missing. This completed the geographical layer of the model and provided the context needed to understand where

Keystra's workforce operates, where HSE events occur, and how safety performance varies across different operating environments.

## 11. DATE DIMENSION DEVELOPMENT

With the organisational, workforce, operational, and geographical structures established, the next step was to introduce a consistent time dimension. This provided a common calendar that could be used across the HSE fact tables and ensured that workforce activity, exposure, incidents, and corrective actions could be analysed consistently over time.

A continuous Dim\_Date table was created covering 1 January 2022 to 31 July 2026, with each calendar date represented as a separate record. Alongside the Date and Date\_ID, the dimension included Year, Quarter, Month Number, Month Name, Month-Year, Week Number, Day Name, Day of Week, and Weekend Indicator. These attributes provide the structure required for chronological reporting and time-based analysis in Power BI, including monthly incident trends, annual exposure patterns, and corrective-action activity.

The completed date dimension was validated by checking the date range, record and column counts, date continuity, unique Date IDs, expected reporting years, and missing dates. These checks confirmed that the calendar was complete and continuous and could serve as the shared time reference across the HSE analytical model.

### INTEGRATED DIMENSION AND LOOKUP VALIDATION

Once all the individual dimensions had been developed, the next step was to make sure that the reference layer worked as a complete structure, rather than simply having a collection of separate tables.

The review covered Dim\_Date, Dim\_Employee, Dim\_BusinessUnit, Dim\_Department, Dim\_Asset, Dim\_JobRole, Dim\_IncidentType, Dim\_Hazard, Dim\_Shift, Dim\_Contractor, and Dim\_Location. The required records, identifiers, and descriptive fields were checked to ensure that each dimension was available and properly structured for use by the fact tables. This was particularly important because the next stages of the project would rely on these dimensions to provide consistent context. Employees needed to be linked to the right organisational structure, assets needed valid locations and business units, incidents needed appropriate incident and hazard classifications, and all transactional records needed a consistent date reference.

With the reference structures confirmed, the dimension layer of the Keystra HSE data warehouse was considered ready. The project could therefore move into the next stage:



generating the transactional data that would capture workforce exposure, HSE incidents, and corrective actions.

## **FACT TABLES DEVELOPMENT**

### **1. FACT\_HSE\_EXPOSURE**

Once the workforce and supporting dimensions were in place, the next step was to capture how much work activity was actually taking place across the organisation. This was the purpose of Fact\_HSE\_Exposure.

Exposure hours are important because incident numbers on their own do not tell the full safety story. The amount of time employees spend working provides the context needed to understand whether incident levels are high or low relative to the amount of work being performed. It also provides the denominator for key HSE measures such as TRIR and LTIFR.

The table was built at the employee-month level, covering January 2022 to July 2026. An Employee × Month framework was created and enriched with each employee's Business Unit, Department, Asset, Shift, Employment Type, and Employment Status. Monthly exposure was generated around a 173.8-hour baseline, with reasonable variation introduced to reflect differences in working patterns. Employee status was also taken into account, so employees who were on leave or otherwise not actively working did not accumulate normal exposure hours. Each reporting month was then linked to Dim\_Date through Date\_ID, making the exposure records ready for time-based analysis.

Before the table was considered complete, it was checked for overall exposure levels, unusual or zero values, missing data, valid date relationships, employee-month coverage, and duplicate records. This left Fact\_HSE\_Exposure as the foundation for measuring workforce activity and, more importantly, provided the context needed to interpret the HSE incidents generated in the next stage.

### **2. FACT\_HSE\_INCIDENT**

Once workforce exposure had been established, the next step was to build Fact\_HSE\_Incident, which captures the HSE events that occurred across Keystra during the reporting period. Each row represents a single incident and connects it to the employee, department, business unit, asset, shift, location, incident type, hazard, and other relevant information.

Rather than assigning incidents completely at random, the generation process was designed to reflect differences in operational risk. An active employee pool was first created, after which department, asset, and shift risk factors were combined to determine

the relative incident propensity of each employee. The resulting probabilities were validated and normalised before being used for weighted sampling. Incident volume was also informed by the workforce exposure profile and baseline event frequency, allowing the number and distribution of incidents to remain connected to the amount of work being performed. Sampling was done with replacement, so an employee could legitimately appear in more than one incident.

The selected employees were then linked back to their organisational and operational attributes, including Business Unit, Department, Asset, Shift, and Work Location. These relationships were checked against the employee dimension to ensure that the incident records remained consistent with the established workforce structure.

The timing of incidents was based on the monthly exposure profile. Incidents were distributed across January 2022 to July 2026, individual dates were generated within the assigned months, and the appropriate Date\_ID was created for connection to the calendar dimension.

The type of incident was then determined using baseline probabilities that were adjusted according to the relevant department and asset. The `get_incident_type_weights()` logic allowed different operational environments to have different incident profiles instead of applying the same distribution everywhere.

Hazards were modelled in a similar way. Department- and asset-specific hazard profiles were developed and tested across offshore, gas processing, maintenance, logistics, and office environments before hazards were assigned to incidents through weighted selection. The resulting distribution was reviewed, and every Hazard ID was checked against Dim\_Hazard.

Because contractors are an important part of activities such as maintenance, engineering, logistics, construction, and specialist technical work, contractor involvement was also modelled. The likelihood of contractor involvement varied according to the department and asset, with the probability capped at 60% to avoid unrealistic results. Contractor-related incidents received a valid Contractor ID, while employee-only incidents were left without one.

The consequences of each incident were then developed through Recordable Incident, LTI, Fatality, and Near Miss flags. Lost workdays were generated according to incident severity, while operational downtime captured the potential disruption to Keystra's activities. This meant the fact table could show not only that an incident occurred, but also how seriously it affected people and operations.

The financial side of each incident was captured through six cost components:

>> Property Damage

>> Environmental Remediation

>> Emergency Response

>> Medical & Compensation

>> Operational Disruption

>> Other Costs

These costs were generated in relation to factors such as severity, downtime, and lost workdays. Keeping the components separate also made it possible to examine where financial losses were coming from and compare incident frequency, total cost, and average cost across incident types. Financial checks were performed to ensure that the cost components remained properly reconciled. Finally, the completed incident fact was validated from several angles. The checks covered record structure, unique Incident IDs, missing values, foreign-key relationships, incident classifications, hazards, contractor references, severity indicators, lost workdays, downtime, and financial impact.

By the end of this stage, Fact\_HSE\_Incident had become more than a list of incidents. It provided a connected picture of who was involved, where and when the event occurred, what type of event and hazard were involved, whether contractors were present, how serious the outcome was, what operational disruption followed, and what financial impact resulted. With the incident records now complete and validated, the next stage was to capture how Keystra responded to those incidents through corrective actions.

### 3. FACT\_HSE\_CORRECTIVEACTION

Following the completion of the incident fact, the next stage focused on how Keystra responds to HSE events. Corrective actions were modelled separately because one incident can require several interventions, such as equipment repair, employee retraining, procedure changes, or engineering improvements.

A dedicated Dim\_CorrectiveAction was first established with eight action types: Engineering Control, Administrative Control, Procedure / Process Revision, Training & Competency Improvement, PPE Improvement, Maintenance / Repair, Inspection / Monitoring, and Emergency Preparedness. Each category was given a unique identifier and validated before being used in the fact table.

The Fact\_HSE\_CorrectiveAction records were then generated directly from validated incidents, creating a realistic one-to-many relationship between incidents and corrective actions. A total of 3,060 corrective actions were created, with each action receiving a unique Corrective Action ID. Each record retained its connection to the originating incident through the Incident ID, Employee ID, Business Unit, Department, Asset, Incident Type, Hazard, and Incident Date.

Management attributes were then added to describe how each action would be handled. These included Corrective Action Type, Priority, Responsible Function, and Investigation Status. Priorities ranged from Critical to Low, while investigation status was represented as Completed, In Progress, or Pending. The distributions were weighted rather than uniform to better reflect how corrective actions would typically be managed in an energy organisation.

The action lifecycle was then modelled using 31 July 2026 as the reporting cutoff. Action dates were generated after the related incidents, target closure periods were assigned according to priority, and due dates were calculated accordingly. Actions were classified as Closed, Open, or Overdue, with closure dates generated only where completion was realistically possible before the reporting cutoff. Days Open was then calculated based on whether an action had already been closed or was still outstanding. To make the lifecycle easier to analyse, Overdue Flag and Closed Flag were also created. This allowed closure and overdue KPIs to be calculated efficiently in Power BI.

The corrective action lifecycle was checked carefully to ensure that the dates followed a sensible sequence: an action could not begin before its incident, a due date could not occur before the action date, and a closure date could not occur before the action began. Negative Days Open, invalid dates, reporting-cutoff violations, and inconsistent status/flag combinations were also checked. The model went further by distinguishing action closure from action effectiveness. Closed actions could be assessed as Effective, Partially Effective, or Ineffective, while Open and Overdue actions were classified as Not Yet Verified. Verification dates were generated for completed actions and checked to ensure that they occurred after closure and within the reporting period.

The financial side of corrective-action management was also incorporated. Costs were influenced by both action type and priority, with engineering and maintenance interventions generally carrying greater costs than administrative or procedural actions. Critical actions received the highest priority adjustment, followed by High, Medium, and Low. Total, average, minimum, and maximum costs were reviewed, alongside cost distributions by priority and corrective-action type.

The final validation brought everything together. The table was checked for its structure, row and column counts, unique Corrective Action IDs, status and priority distributions, closure rate, overdue rate, Days Open, high-priority outstanding actions, referential integrity, business rules, flags, effectiveness, and financial values. Foreign keys were also checked against the relevant incident and dimension tables to ensure that no orphan records were introduced. Additional spot checks were performed for potentially problematic records, including actions with dates after the reporting cutoff, closure dates before action dates, and action dates occurring before the related incident.

With these checks completed, Fact\_HSE\_CorrectiveAction was confirmed as a validated part of the warehouse. At this point, the three core HSE fact areas—workforce exposure, HSE incidents, and corrective actions—had been established. The next stage could therefore move from validating individual tables to checking whether they worked together consistently as one integrated HSE analytical warehouse.

## FINAL DATA OVERVIEW

The completed Keystra Energy Group Enterprise Workforce Safety Intelligence warehouse consists of 12 dimension tables and 3 fact tables, giving 15 datasets in total. Together, these tables provide a structured view of Keystra’s workforce, organisational structure, operational assets, geographical footprint, workforce exposure, HSE events, and corrective-action activities. The analytical period covers 1 January 2022 to 31 July 2026, with 31 July 2026 used as the reporting reference date. The workforce is represented by 4,000 employees, while the fact tables capture 220,000 exposure records, 6,370 HSE incidents, and 3,060 corrective actions.

| DATASET              | ROWS  | COLUMNS | PURPOSE                                              |
|----------------------|-------|---------|------------------------------------------------------|
| Dim_Asset            | 13    | 3       | Defines Keystra’s operational assets                 |
| Dim_BusinessUnit     | 5     | 2       | Defines the five business units                      |
| Dim_Contractor       | 8     | 4       | Provides the contractor reference                    |
| Dim_CorrectiveAction | 8     | 2       | Classifies corrective action types                   |
| Dim_Date             | 1,673 | 11      | Provides the shared calendar and reporting structure |
| Dim_Department       | 7     | 2       | Defines the organisational departments               |
| Dim_Employee         | 4,000 | 25      | Provides the workforce master records                |
| Dim_Hazard           | 12    | 3       | Defines the hazard categories used in HSE analysis   |
| Dim_IncidentType     | 10    | 3       | Defines the HSE incidents classification             |
| Dim_JobRole          | 35    | 3       | Defines the workforce job role structure             |
| Dim_Location         | 8     | 5       | Provides geographical and operational context        |

|                                   |                |           |                                                         |
|-----------------------------------|----------------|-----------|---------------------------------------------------------|
| <b>Dim_Shift</b>                  | <b>3</b>       | <b>2</b>  | <b>Defines workforce shift patterns</b>                 |
| <b>Fact_HSE_Exposure</b>          | <b>220,000</b> | <b>13</b> | <b>Captures monthly employee exposure</b>               |
| <b>Fact_HSE_Incident</b>          | <b>6,370</b>   | <b>27</b> | <b>Captures HSE incidents and their consequences</b>    |
| <b>Fact_HSE_CorrectiveActions</b> | <b>3,060</b>   | <b>24</b> | <b>Captures actions taken in response to HSE events</b> |

The 12 dimension tables provide the context needed to interpret the transactional data. They describe who works within Keystra, where they work, what roles they perform, which assets and business units they support, when activities occur, and how HSE events, hazards, contractors, and corrective actions are classified.

The three fact tables capture the main HSE activities and events within that structure.

**Fact\_HSE\_Exposure** contains 220,000 employee month records and represents the amount of workforce exposure accumulated throughout the reporting period. It provides the activity denominator needed for measures such as TRIR and LTIFR, while also supporting analysis of exposure across business units, departments, assets, shifts, and employment groups.

**Fact\_HSE\_Incident** contains 6,370 individual HSE events. Beyond identifying where and when incidents occurred, it captures their classification, hazard, workforce context, severity, lost workdays, operational downtime, and financial consequences.

**Fact\_HSE\_CorrectiveAction** contains 3,060 actions linked back to the incidents that generated them. The table captures the organisation's response through action type, priority, responsible function, status, closure timeline, effectiveness, and cost.

Taken together, the warehouse creates a connected analytical flow from:

**Workforce → Exposure → HSE Incident → Consequence → Corrective Action**

This structure allows the analysis to move beyond simply asking how many incidents occurred. It supports a broader view of HSE performance by connecting the level of workforce activity to incident frequency, examining where risks are concentrated, understanding the consequences of HSE events, assessing their operational and financial impact, and evaluating how effectively identified risks are addressed. The completed warehouse therefore provides the data foundation required for the next stage of the project: Power BI data modelling, DAX measure development, dashboard design, and enterprise HSE performance analysis.

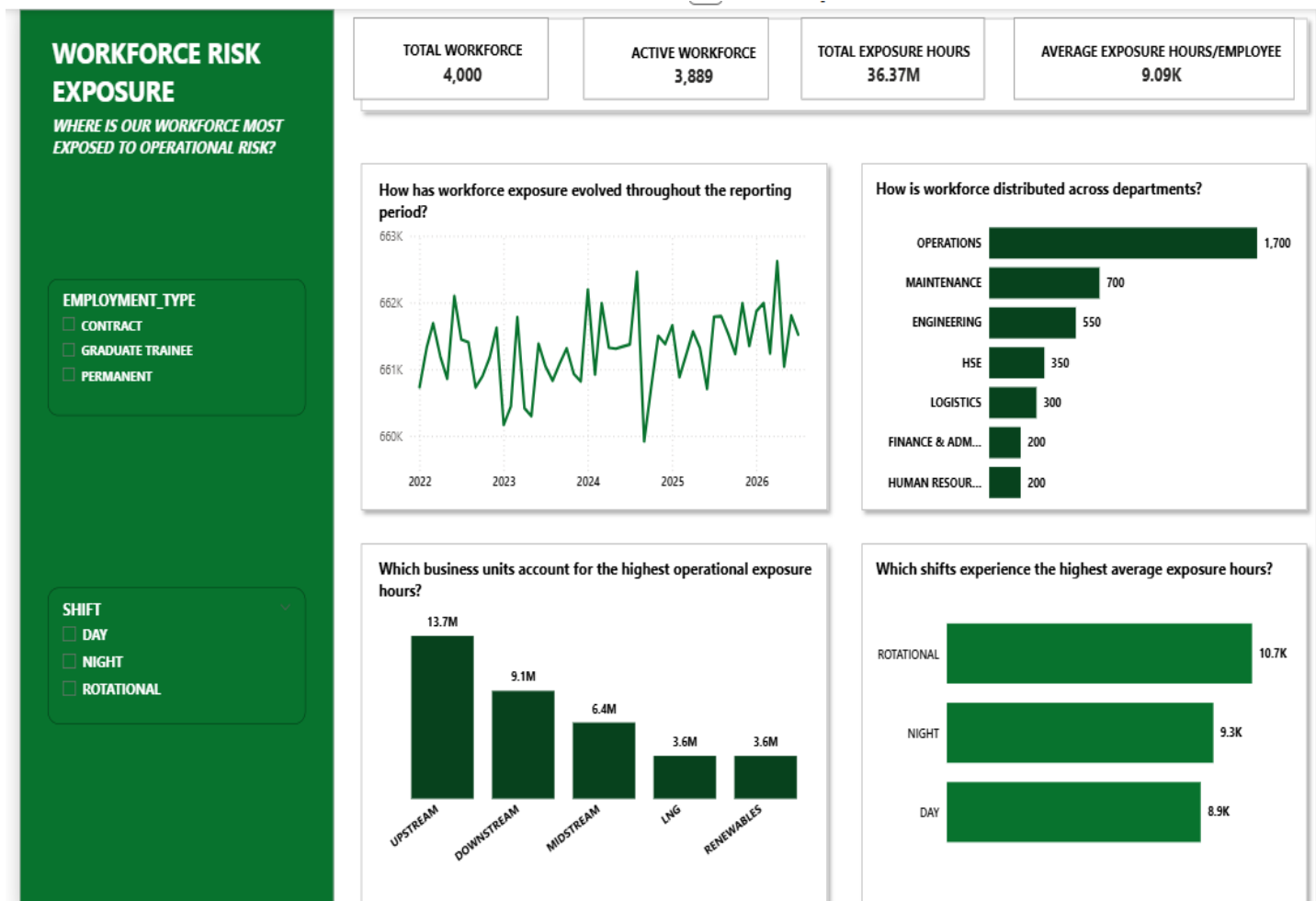
With the HSE data warehouse completed, the final stage of the project focused on evaluating Keystra's HSE performance through an interactive Power BI dashboard. This stage shifts the focus from data preparation to understanding what the data reveals about the organisation's safety performance.

For an integrated energy company like Keystra, HSE performance cannot be assessed by incident counts alone. Management also needs to understand where the workforce is most exposed to operational activities, how safety performance varies across business units and assets, which hazards contribute most to incidents, and how effectively corrective actions are addressing identified risks.

To support these objectives, the dashboard was organised into three interconnected pages. The first page provides an overview of the workforce and exposure profile, establishing the operational context for HSE performance. The second page examines operational safety performance through incident trends, hazard patterns, and key performance indicators. The final page focuses on the consequences of those incidents by analysing operational impact, financial costs, and corrective action performance.

Together, these dashboards provide a clear view of Keystra's HSE performance, enabling management to identify areas of elevated risk, monitor safety outcomes, evaluate corrective actions, and make informed decisions to strengthen workforce safety across the organisation.

## WORKFORCE RISK EXPOSURE



Before looking at whether Keystra is performing well from an HSE perspective, I first needed to understand the environment in which those safety outcomes were occurring. The reason is simple: an incident count means very little without knowing how much work is being carried out and where that work is concentrated. A business unit with a much larger workforce and substantially more operating hours will naturally have more opportunities for incidents than a smaller business unit. If I were to compare them using incident counts alone, I could easily conclude that the business unit with the most incidents is the least safe, when the real difference may simply be the amount of work being performed.

I therefore used the first page to establish Keystra's workforce and exposure baseline; where people are concentrated, where the greatest amount of operational activity is taking place, and how exposure differs across the workforce.

The workforce is concentrated mainly within Operations, Maintenance and Engineering. This is consistent with Keystra's operating environment because these functions are closely tied to production, asset maintenance and the technical activities required to keep the business running.

The exposure profile tells an even more important story. Upstream accounts for the largest operational exposure at approximately 13.7 million hours, followed by Downstream at 9.1 million and Midstream at 6.4 million hours, while LNG and Renewables operate at considerably lower exposure volumes. That immediately gave me something important to consider when I moved into the incident analysis: "Upstream having more incidents does not automatically mean that Upstream is performing worse. It is also carrying significantly more operational activity". This is why I did not want to use incident volume as the only measure of safety performance. I needed to know what was happening relative to the amount of work being performed.

The shift profile added another piece to that picture. Employees working rotational shifts have the highest average exposure hours per employee, followed by night and day shifts. For Keystra, this is relevant because its operations include work environments where employees may spend extended periods working around operational assets and production activities. The difference in exposure across shifts means that shift pattern is worth considering when interpreting subsequent incident results.

At this point, the purpose of Page 1 was not to decide which part of Keystra was safest or least safe. It was to establish the context needed to make that judgement properly. The main question coming out of this analysis was therefore:

*"Given the amount and type of work being performed across Keystra, where are safety outcomes occurring at a level that deserves attention?"*

This question takes us into the second page.



## **OPERATIONAL SAFETY PERFORMANCE**

Once the workforce and exposure baseline was established, I could look at Keystra's actual safety outcomes with much more context. Across the reporting period, Keystra recorded 6,370 HSE incidents, including 702 recordable incidents and 3,710 near misses, with a TRIF of 3.86 and LTIFR of 0.41. There were also two fatalities.

But the first thing I had to be careful about was not interpreting these numbers in isolation. Upstream, for example, records the highest incident volume, but Page 1 has already shown that it also carries the highest exposure at approximately 13.7 million hours. So simply saying "Upstream has the most incidents" would not tell management whether its safety performance is actually worse. This is where the rates become more useful.

When the analysis was broken down by department, Logistics recorded the highest TRIF at 4.7, followed by Operations at 4.2 and Engineering at 4.1. This changes the picture from what we might assume by looking at workforce size alone. Operations has the largest employee population, yet Logistics has the highest injury frequency relative to exposure. That makes Logistics worth investigating further, the question is no longer simply how many incidents occurred there, but what about the work being performed in Logistics is producing a higher injury rate relative to exposure?

From there, I looked at what was actually driving the incidents. The five major hazards contributing most to incident activity were Vehicle/Mobile Equipment, Lifting and Material Handling, Ergonomic/Manual Handling, Dropped Objects and Hydrocarbon Release. However, I did not want to treat these hazards as though they represented the same problem.

When I looked more closely at the incident profile, another distinction emerged. For overall incident frequency, Hydrocarbon Release, Pressure/Stored Energy and Slips, Trips and Falls were among the leading hazards. But when I narrowed the analysis to recordable incidents, the picture shifted towards Vehicle/Mobile Equipment and Lifting and Material Handling. Vehicle/Mobile Equipment accounted for 48 incidents associated with Restricted Work Cases, while Lifting and Material Handling accounted for 25 Medical Treatment Cases. That difference matters because it tells us that the hazards occurring most frequently are not necessarily the hazards producing the greatest recordable injury outcomes. For Keystra, that means the response cannot simply be based on whichever hazard has the biggest incident count. The organisation needs to understand both frequency and consequence.

I then looked at the circumstances surrounding the major incident driving hazards particularly Slips, Trips and Falls, Pressure/Stored Energy, Hydrocarbon Release and Working at Height. These incidents were predominantly associated with the Day Shift, permanent employees and the Operations department. I would not interpret this as

meaning that day-shift workers or permanent employees are inherently more exposed to risk. Rather, it tells us where the incident population is concentrated. That gives Keystra a more focused starting point for asking what is happening within those operational activities, how work is being supervised and whether there are common conditions or control weaknesses behind the events.

The asset level analysis then gave the findings even more context. Offshore Platform A, for example had Hydrocarbon Release as its leading hazard with 32 incidents, followed by Working at Height with 25, and Fire and Explosion with 23. But again, the incident count alone does not tell the whole story. Hydrocarbon Release was primarily associated with First Aid Cases, while Working at Height and Fire/Explosion appeared strongly among Near Misses. This tells us that the hazards are producing different outcomes and should not automatically be treated as equivalent simply because their incident counts are high. Hydrocarbon Release remains particularly important because of the nature of the hazard. A First Aid Case may represent a relatively limited immediate outcome, but a hydrocarbon release can have much greater escalation potential in an energy operating environment.

Then came the finding that changed the severity picture completely. The fatality identified in the data was associated with a Dropped Object incident on Offshore Platform B during the Day Shift. This was important because Dropped Objects were not necessarily the highest frequency hazard across the entire dataset. Yet this hazard had already demonstrated that it could result in the most serious possible consequence.

That led to an important distinction in how I interpreted the safety profile:

*A hazard does not have to be the most frequent hazard to be a major management concern.*

Vehicle/Mobile Equipment and Lifting/Material Handling require strong preventive attention because they contribute significantly to recordable injury activity. Hydrocarbon Release requires continued process safety attention because of its escalation potential. Dropped Objects require particularly strong control assurance because the data shows that this hazard has already resulted in a fatality. So rather than trying to identify one single “worst” hazard, the analysis separates what is happening frequently from what could cause the greatest harm.

And this is where the first two pages come together.

Page 1 showed that Keystra is not operating at the same level of exposure across its businesses. Upstream carries considerably more operational exposure than the other business units.

Page 2 then showed what is happening within that exposure: some departments have higher injury rates, some hazards occur more frequently, some hazards generate more recordable injuries, and some less frequent hazards have demonstrated the potential for

severe consequences. That means Keystra cannot rely on one measure to decide where to intervene. The organisation needs to look at exposure, frequency and consequence together.

There is still one important question left which is knowing where incidents are occurring, what hazards are involved, and where those safety issues are creating the greatest cost to the business,

*how much operational time is being lost?*

*how many workdays are being lost?*

*are the risks identified being followed through with effective corrective action?*

That is where Page 3 takes the analysis, it moves from;

*Where is Keystra's exposure?*

to:

*What is happening within that exposure?*

and finally,

*Where is that safety performance becoming a business problem, and where does Keystra need to intervene?*

## OPERATIONAL SAFETY PERFORMANCE

WHERE ARE SAFETY INCIDENTS OCCURRING?

TOTAL INCIDENTS  
6370

TOTAL RECORDABLE INCIDENTS  
702

NEAR MISSES  
3710

TRIF  
3.86

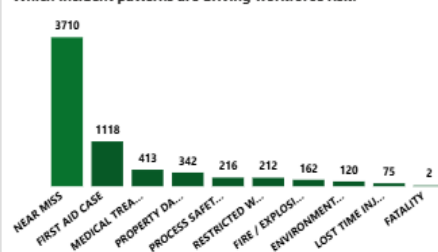
LTIRF  
0.41

FATALITIES  
2

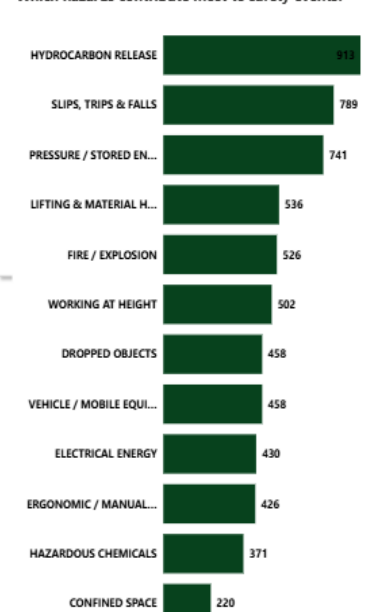
How is incident frequency changing over time?



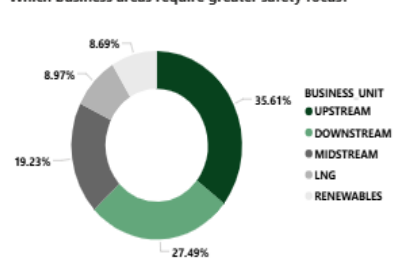
Which incident patterns are driving workforce risk?



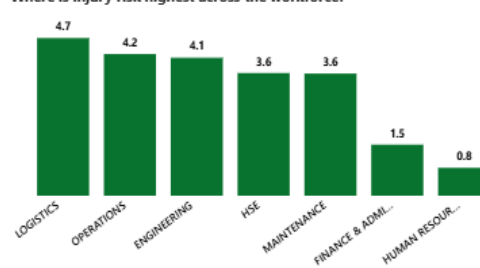
Which hazards contribute most to safety events?



Which business areas require greater safety focus?



Where is injury risk highest across the workforce?



## BUSINESS IMPACT AND CORRECTIVE ACTION

*Where are safety issues having the biggest impact on Keystra, and is the organisation responding effectively?*



After identifying where incidents are occurring and what hazards are driving them, the next step was to understand what those incidents are actually costing Keystra. An incident does not only show up as a number in an HSE register. Depending on its severity, it can take equipment out of service, keep employees away from work, create financial losses and generate corrective actions that still require attention long after the incident itself has occurred. For this reason, I looked beyond incident counts and TRIF to understand the operational, workforce and financial consequences associated with different assets. I then compared these consequences with the corrective actions raised against them. This was important because the asset with the highest safety risk may not necessarily be the asset creating the greatest business impact, and that is exactly what the analysis showed.

Different assets are creating different kinds of problems, the asset level analysis highlighted three assets that stood out, but for very different reasons.

### Gas Processing Plant

The Gas Processing Plant emerges as an asset that deserves closer attention. Its TRIF of 6.02 is the highest across the assets yet it does not lead in incident volume, downtime or incident cost. It recorded 319 incidents and 49 recordable incidents, resulting in 6,362 downtime hours and 1,022 lost workdays. The asset also incurred approximately 9.40M in direct incident costs, alongside 6.65M in corrective action costs, while 22 corrective actions remained open and 14 were overdue.

This combination is what makes the Gas Processing Plant stand out. The concern is not simply the number of events occurring at the asset but because its recordable injury rate is high relative to the level of workforce exposure, suggesting that something about the nature of the work, the hazards encountered, or the conditions under which the work is being performed may warrant further attention.

The deeper analysis begins to show where that attention may be directed; Lifting and material handling stands out among the hazards associated with recordable incidents and lost workdays, while vehicle and mobile equipment also features prominently. There is also a notable workforce pattern: contract employees working rotational shifts recorded the highest TRIF at 18.08, whereas permanent employees and day-shift workers accounted for the highest lost workdays.

Taken together these findings make the Gas Processing Plant more than an asset with a high headline TRIF, but points to specific areas of exposure and consequence that management should examine more closely. The presence of open and overdue corrective actions makes this particularly relevant. The question is whether the risks reflected in the incident data are already being addressed adequately, or whether some of the underlying exposures remain insufficiently controlled.

At this stage, the analysis is not intended to claim that a single hazard, shift or employee group is the definitive cause of the Gas Processing Plant's higher injury rate. Rather, it identifies where the evidence is strong enough to warrant management attention and further investigation. This is the purpose of moving from the incident analysis into the intervention focused view, that is to determine which risks carry the greatest consequence, where unresolved actions may be leaving exposure insufficiently controlled, and where Keystra's management attention is most needed.

#### **Offshore Platform A:**

Offshore Platform A emerges as the asset where HSE performance is translating most clearly into operational disruption, workforce impact and financial exposure. The platform recorded 963 incidents and 104 recordable incidents, alongside 14,459 downtime hours and 2,985 lost workdays. It also recorded approximately ₦23.05 million in direct incident costs which is the highest across the assets.

Beyond the cost of incidents themselves, corrective actions associated with the platform represent a further ~~N~~22.17 million, with 89 actions still open and 72 overdue.

These figures establish the scale of the impact, but they do not by themselves show where management attention should be directed. The analysis therefore goes further by examining the hazards, incident types, workforce consequences, financial exposure, and corrective action position to identify where the impact is most concentrated.

The downtime analysis provides a clear indication of where HSE events are most strongly associated with operational disruption. Dropped object hazards account for 5,502 downtime hours making them the largest contributor. They are followed by ergonomic/manual handling hazards at 4,294 hours, and vehicle/mobile-equipment hazards at 2,954 hours. When downtime is examined by incident type, Property Damage, and Process Safety Events emerge as the two categories associated with the greatest downtime. This shifts the interpretation of Offshore Platform A from simply being an asset with a high volume of incidents to one where specific hazards and incident types are associated with substantial operational disruption. Dropped objects are particularly important because they contribute significantly to downtime and also feature among the leading contributors to incident cost. The concentration of impact around these risk areas provides a more actionable basis for management intervention than incident volume alone.

The workforce impact reveals another dimension of the problem. Medical Treatment Cases account for 1,497 lost workdays, the highest among the incident categories, while Restricted Work Cases account for 48 recordable incidents, the highest count in that category. When the hazards associated with these outcomes are examined, vehicle/mobile-equipment, and lifting/material-handling hazards feature prominently across both recordable incidents and lost workdays. This indicates that the HSE impact at Offshore Platform A extends beyond equipment and production. It is also associated with lost workforce capacity and reduced availability, creating an additional operational burden for the asset. The result is a broader business impact: people are being affected, work is being disrupted and financial costs are being incurred.

The financial analysis provides further evidence of where the impact is concentrated. Property Damage is the leading incident type by total incident cost contributing approximately ~~N~~4.56 million. At the hazard level, dropped-object and ergonomic/manual-handling hazards are among the leading contributors to total incident cost. The significance is not simply the size of the financial figures. It is the recurrence of particular risk areas across multiple measures of impact. Where the same hazards appear across downtime and financial exposure, they become stronger candidates for management attention. This is particularly relevant for dropped objects which feature prominently in both operational disruption and incident cost.

The incident and impact profile becomes more significant when considered alongside the corrective action position. Offshore Platform A has 89 open corrective actions and 72 overdue actions, indicating that a substantial number of identified issues have yet to reach closure. More importantly, the backlog includes 3 Critical and 20 High-priority open actions, together with a further 4 Critical and 14 High-priority overdue actions. This makes the corrective action position more significant than simply having a large number of outstanding actions.

The presence of Critical and High-priority actions that remain unresolved or overdue indicates that some higher severity issues still require management attention. Hydrocarbon Release provides a notable example of this unresolved action pattern. It has the highest number of closed corrective actions at 47, indicating that considerable corrective effort has already been directed toward the hazard. However, it also has the highest number of open actions at 13 and overdue actions at 12. This suggests that although significant action has already been taken, a substantial amount of corrective work remains outstanding. The focus should therefore extend beyond the number of actions completed to whether the controls and corrective measures are achieving their intended outcomes.

The effectiveness analysis reinforces this point. Hydrocarbon Release has 25 actions classified as Not Yet Verified and 16 classified as Partially Effective, while Electrical Energy and Working at Height each have 4 actions classified as Ineffective. This distinction is important because action closure does not necessarily mean that the underlying risk has been effectively addressed. Some completed actions have not yet been verified for effectiveness, while others have only achieved partial effectiveness or have been classified as ineffective.

#### *So what does the analysis tells management?*

Taken together, the evidence places Offshore Platform A in a distinct position. The asset is not simply experiencing a high number of HSE events; its HSE profile is associated with greater consequences across operational continuity, workforce capacity and financial performance. The key signals are:

- Operational impact: 14,459 downtime hours, with dropped objects, ergonomic/manual handling and vehicle/mobile equipment among the leading hazard contributors.
- Workforce impact: 2,985 lost workdays, with Medical Treatment Cases, vehicle/mobile equipment and lifting/material handling featuring prominently in the workforce impact.
- Financial impact: ~~£~~23.05 million in direct incident costs, with Property Damage leading incident-type costs and dropped object and ergonomic/manual-handling hazards among the leading cost contributors.

- Control and remediation gap: 89 open and 72 overdue corrective actions, including unresolved Critical and High-priority actions and notable concerns around corrective-action effectiveness.

The strongest management signal is the recurrence of specific risk areas across multiple dimensions of impact. Rather than treating each KPI independently, the analysis points toward a more concentrated set of concerns: dropped objects, ergonomic/manual handling, vehicle/mobile equipment, lifting/material handling, Property Damage, Process Safety Events and unresolved corrective actions particularly those associated with Hydrocarbon Release.

Importantly, the dashboard does not establish causation. The data shows that these hazards and incident types are associated with high levels of downtime, workforce impact and financial exposure, but it does not prove that any individual hazard or overdue corrective action directly caused a specific loss. Establishing causality would require deeper operational investigation into individual events, underlying conditions, control failures, and the effectiveness of controls in the field.

What the analysis does provide is a focused starting point for that investigation. The central question is no longer simply:

*“Why does Offshore Platform A have a high HSE impact?”*

The analysis allows management to ask a much more targeted question:

*Why are these specific hazards and incident types associated with such significant operational, workforce and financial consequences at Offshore Platform A, and are the controls and corrective actions in place strong enough to prevent these risks from recurring?*

Management intervention should therefore begin with the high consequence risk areas, particularly those that appear repeatedly across downtime, workforce impact and cost. This should be supported by a review of the effectiveness of existing controls, outstanding corrective actions, and the reasons Critical and High-priority actions remain open or overdue.

Offshore Platform A should therefore be treated as a clear management priority, not simply because it has high HSE numbers, but because the data identifies specific risk areas, significant business consequences and unresolved control concerns that warrant deeper operational investigation.

#### **Refinery Complex:**

Refinery Complex presents a different HSE challenge from Offshore Platform A. While Offshore Platform A emerged as the asset with the greatest operational and financial



impact, Refinery Complex recorded the highest workforce consequence with 3,690 lost workdays. The asset recorded 872 incidents, 94 recordable incidents and a TRIF of 3.91, alongside 12,334 downtime hours and approximately ₦19.82 million in direct incident costs. Its corrective-action profile also shows a significant outstanding burden, with 76 open actions and 57 overdue actions, alongside approximately ₦18.09 million in corrective-action costs.

The significance of Refinery Complex therefore does not come from having the highest injury frequency or the greatest operational disruption, it comes from the extent to which HSE events are affecting workforce availability. This raised a more specific question:

*What is driving the 3,690 lost workdays at Refinery Complex?*

The analysis was drilled down from the asset level into incident type. Medical Treatment Cases account for 2,936 lost workdays, representing approximately 79.6% of the asset's total lost workdays. They also have the highest recordable incident count, with 61 cases. This shows that the workforce consequence is not evenly distributed across incident types. It is heavily concentrated within Medical Treatment Cases.

The analysis was then taken one level deeper to identify the hazards associated with these cases. Two hazards dominate the result:

- \* Vehicle & Mobile Equipment — 1,905 lost workdays
- \* Lifting & Material Handling — 1,026 lost workdays

Together, these account for 2,931 lost workdays approximately 99.8% of the lost workdays associated with Medical Treatment Cases. This changes the interpretation of Refinery Complex's performance. The issue is not simply that the asset has accumulated the highest number of lost workdays, but that the workforce impact is highly concentrated around Medical Treatment Cases, and within those cases overwhelmingly around Vehicle & Mobile Equipment and Lifting & Material Handling.

The next question was whether the corrective action data showed unresolved exposure around these same areas, and it does. Vehicle & Mobile Equipment, which has the highest workforce-impact contribution at 1,905 lost workdays, has Critical and High-priority corrective actions that are overdue. Lifting & Material Handling, associated with a further 1,026 lost workdays, has Critical and High-priority corrective actions that remain open.

This creates an important distinction between the two hazards. Vehicle & Mobile Equipment represents the stronger immediate concern because it combines the largest workforce consequence with overdue higher-priority corrective actions. Lifting & Material Handling also warrants attention because of its substantial workforce impact and unresolved Critical/High-priority actions.

A separate signal emerged from the broader corrective-action population. Slip, Trips & Falls has the highest overall corrective-action backlog, with 13 open and 10 overdue actions. However, this should not be interpreted as evidence that Slip, Trips & Falls is responsible for the highest lost-workday burden. The analysis is showing two different dimensions of risk: Workforce consequence is concentrated around Vehicle & Mobile Equipment and Lifting & Material Handling. Corrective action backlog is greatest around Slip, Trips & Falls. That distinction matters because the hazard with the largest number of outstanding actions is not necessarily the hazard producing the greatest business consequence.

Taken together, the analysis provides a much clearer picture of Refinery Complex. The asset has the highest workforce consequence across the assets, and that consequence is heavily concentrated around Medical Treatment Cases associated with Vehicle & Mobile Equipment and Lifting & Material Handling. At the same time, higher-priority corrective actions remain unresolved in both areas, with the Vehicle & Mobile Equipment actions already overdue.

The data does not establish that these unresolved actions caused the lost workdays, nor that closing them would automatically prevent future incidents. Establishing that relationship would require further investigation of individual events, operating conditions, control failures and the effectiveness of implemented controls. What the dashboard does provide is a focused basis for management intervention:

*Why are Vehicle & Mobile Equipment and Lifting & Material Handling incidents resulting in such significant workforce absence at Refinery Complex, and are the controls and corrective actions addressing these risks being implemented effectively and on time?*

That is the key HSE question emerging from the Refinery Complex analysis. The strength of the dashboard is not in any single KPI. It comes from the way the three pages progressively move from exposure, to events, to consequence and action.

Page 1 establishes the operating context: how much workforce exposure Keystra is carrying and where that exposure is concentrated.

Page 2 examines what is happening within that exposure: where incidents occur, how injury frequency varies, which hazards and incident types are involved, and where serious consequences are emerging.

Page 3 takes the analysis further by asking what those HSE issues are costing the business and where significant risk remains unresolved.

When the pages are considered together, the analysis shows that there is no single “worst” asset. The answer depends on the dimension of performance being considered.

**Gas Processing Plant has the highest injury frequency, with a TRIF of 6.02.**

**Offshore Platform A has the greatest operational and financial consequence, with the highest downtime and direct incident cost, alongside substantial workforce impact and an unresolved corrective-action burden.**

**Refinery Complex has the greatest workforce consequence, with 3,690 lost workdays.**

**These are not competing conclusions. They describe different ways in which HSE performance is affecting the business.**

**For Gas Processing Plant, the management question is:**

***What is driving the elevated injury frequency relative to exposure?***

**For Offshore Platform A, the question becomes:**

***Why are HSE events producing such significant operational disruption and financial loss, and are the unresolved high-priority risks being effectively addressed?***

**For Refinery Complex, the analysis has narrowed the question further:**

***Why are Medical Treatment Cases particularly those associated with Vehicle & Mobile Equipment, and Lifting & Material Handling resulting in such significant workforce absence?***

**The corrective action analysis then adds another dimension: what is happening after risks have been identified?**

**At Offshore Platform A, Hydrocarbon Release is particularly important because it emerged both as a leading incident hazard, with 32 incidents, and among the hazards with the greatest number of unresolved Critical/High-priority corrective actions, alongside Working at Height.**

**At Refinery Complex, the pattern is different. Vehicle & Mobile Equipment and Lifting & Material Handling dominate the workforce consequence, while Slip, Trips & Falls has the largest overall open and overdue corrective-action population.**

**This difference is important. It shows why management should not assume that the hazard with the most outstanding actions is automatically the hazard creating the greatest business consequence. The analysis also reveals a recurring signal that warrants further investigation: Vehicle & Mobile Equipment appears prominently in both Offshore Platform A and Refinery Complex.**

**At Offshore Platform A, it contributes 2,954 downtime hours.**

**At Refinery Complex, it is associated with 1,905 lost workdays and has overdue Critical/High-priority corrective actions.**

The data does not establish that the underlying causes are the same, nor does it establish a common root cause across the assets. However, the recurrence is sufficiently notable to justify a cross asset investigation into whether common weaknesses exist in areas such as vehicle and equipment management, operating practices, competency, supervision or control effectiveness. This is where the analysis moves beyond describing individual assets and begins to generate enterprise-level learning.

The Dropped Object fatality at Offshore Platform B reinforces the same principle from another angle. The event demonstrates that HSE risk cannot be judged by frequency alone. A single high-consequence event can carry greater significance than a large number of lower-severity incidents.

The corrective-action analysis also showed why closure should not be treated as proof of risk reduction. Within the filtered population associated with the fatal event, 28 actions were initially classified as closed. After effectiveness was considered, only 18 remained classified as effectively closed, and applying the priority filter left only one Critical action in that view. The lesson is not that closed actions are ineffective. It is that administrative closure and demonstrated control effectiveness are different measures.

For significant events and high-consequence hazards, Keystra needs to know not only whether an action has been completed, but whether the control was implemented as intended and whether there is evidence that it is actually reducing risk.

Taken together, the three pages therefore provide Keystra with a structured management framework:

Page 1 — Where is the workforce exposed?

Page 2 — What is happening within that exposure?

Page 3 — What are those events costing the business, and where does unresolved risk remain?

The resulting priorities are different by asset:

**Gas Processing Plant:** investigate the drivers of elevated injury frequency.

**Offshore Platform A:** address the operational and financial consequences of HSE events and examine unresolved high-priority risks, particularly around Hydrocarbon Release and Working at Height.

**Refinery Complex:** investigate the workforce consequences associated with Medical Treatment Cases, particularly Vehicle & Mobile Equipment and Lifting & Material Handling, while addressing unresolved high-priority corrective actions.

**Across Keystra:**

determine whether recurring patterns such as Vehicle & Mobile Equipment point to broader control weaknesses that can be addressed through cross-asset learning.

Most importantly, the analysis changes the management conversation, instead of asking only:

*“How many incidents did we have?”*

Keystra can now ask:

*“Where is HSE risk creating the greatest consequence, what is driving that consequence, where does unresolved exposure remain, and what evidence tells us where management should intervene?”*

That is the shift from HSE reporting to HSE intelligence: using operational data not simply to describe what has happened, but to identify where the business should investigate, prioritise and act.

## CONCLUSION

The analysis gives Keystra a clearer idea of where attention is needed and why. It also shows that the right intervention is not necessarily the same everywhere. Some areas require attention because injuries are occurring more frequently, others because the consequences are more severe, while some require closer follow-up because significant corrective actions remain unresolved.

But there is another question that comes out of this analysis; We have largely been looking at events after they have happened, incidents, injuries, downtime, lost workdays, costs and corrective actions. That tells Keystra where things have gone wrong, but it does not necessarily tell us whether an asset was showing signs of deterioration before the problem appeared.

That is where the analysis naturally leads to the next solution. If an asset is beginning to deteriorate, could Keystra recognise those warning signs early enough to act before they become equipment failure, unplanned downtime or a wider operational and safety problem?

The next solution takes that question forward by shifting from understanding what has happened to identifying what may happen next through predictive asset intelligence.